

NOZZLES

- *Nozzle is duct of smoothly varying cross-sectional area in which a steadily flowing fluid can be made to accelerate by a pressure drop along the duct.*

STEAM NOZZLES

- Nozzle is a duct where the velocity of a fluid increases at the expense of pressure drop. If the fluid is steam, then the nozzle is called as Steam nozzle.
- The flow of steam through nozzles may be taken as adiabatic expansion. The steam possesses a very high velocity at the end of the expansion, and the enthalpy decreases as expansion occurs.
- The major function of nozzle is to produce steam jet with high velocity to drive a turbine.

TYPES OF NOZZLES

- There are three types of nozzles
 1. Convergent nozzle
 2. Divergent nozzle
 3. Convergent-divergent nozzle

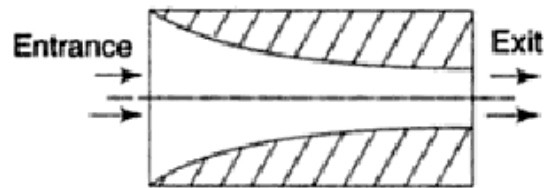


Fig1: Convergent Nozzle

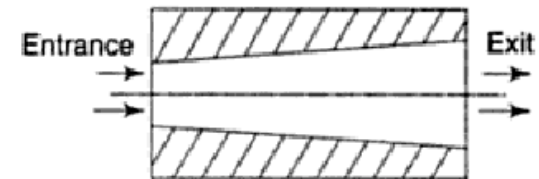


Fig2: Divergent Nozzle

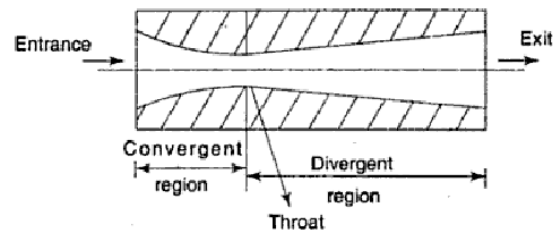
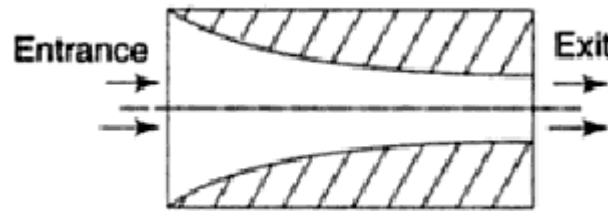


Fig3: Convergent-Divergent Nozzle

1. Convergent Nozzle

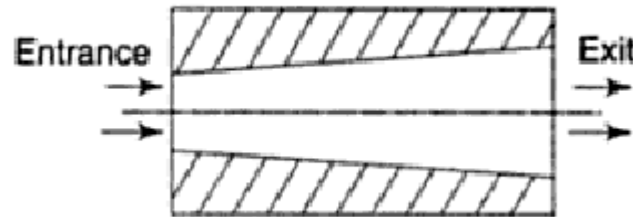
- A typical convergent nozzle is shown in the Fig.1.
- In a convergent nozzle, the cross sectional area decreases continuously from its entrance to exit.



Convergent nozzle

2.Divergent Nozzle

- The cross sectional area of divergent nozzle increases continuously from its entrance to exit.



Divergent nozzle

3. Convergent-Divergent Nozzle

- In this condition, the cross sectional area first decreases from its entrance to the throat and then again increases from throat to the exit.
- This case is used in the case where the back pressure is less than the critical pressure.
- It is widely used in many types of steam turbines.

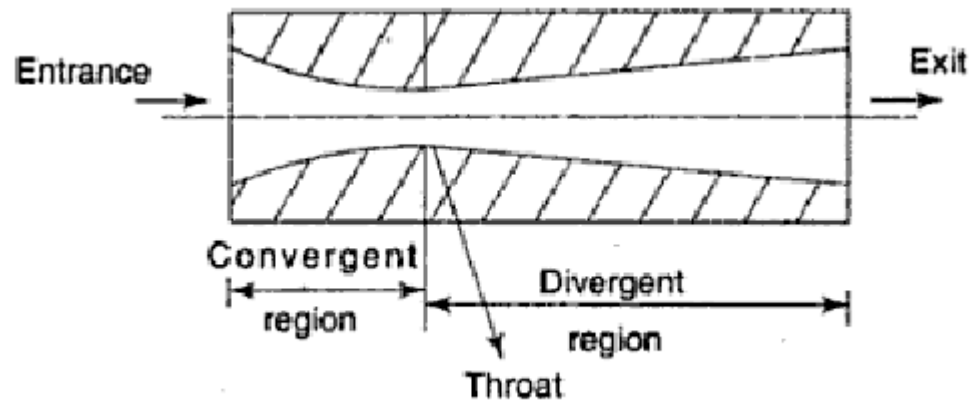
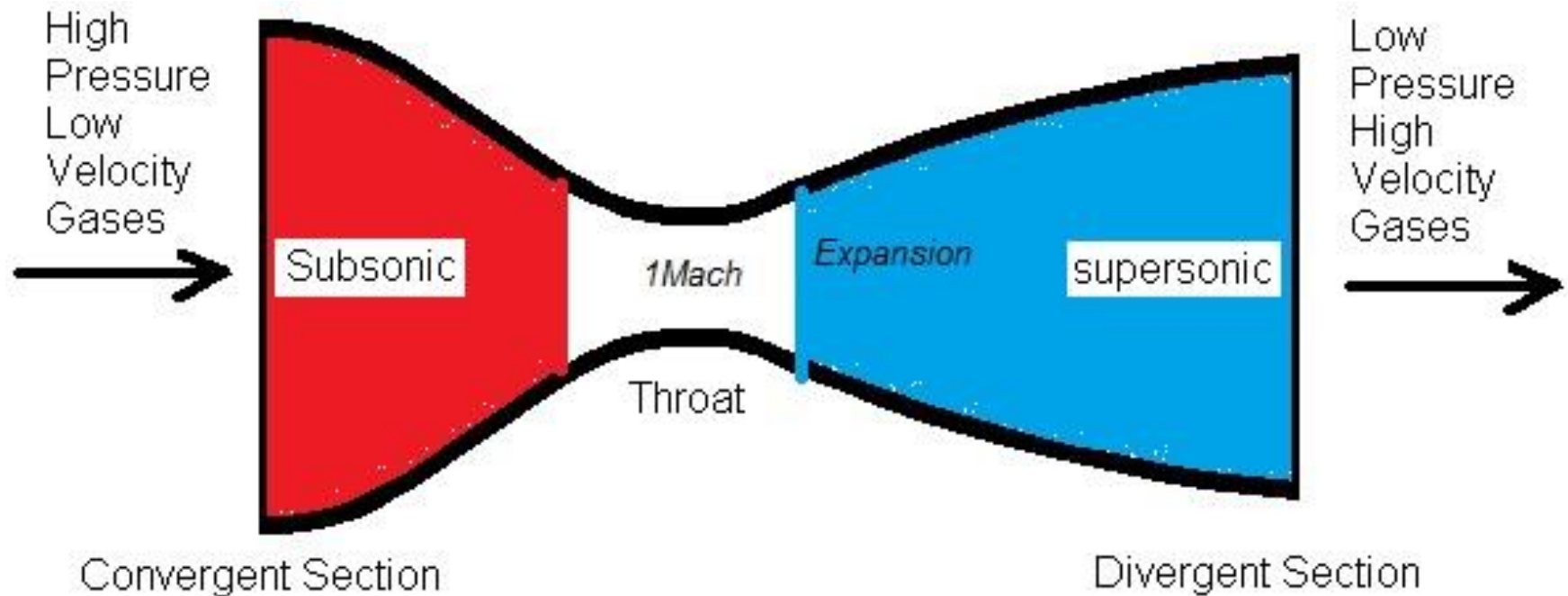


Fig.3. convergent-divergent nozzle

TYPES OF NOZZLES



STEAM NOZZLES

- Friction exists between the steam and the sides of the nozzle; heat is produced as the result of the resistance to the flow.
- A fluid is called compressible if its density changes with the change in pressure brought about by the flow.
- If the density changes very little or does not change, the fluid is said to be incompressible.
- Generally the gases and vapours are compressible, whereas liquids are incompressible.

Equation of Continuity

- Let

- m – mass of fluid in kg/s
- A - cross sectional area m^2
- v – Specific Volume of the fluid m^3/kg
- V – velocity across the section m/s

$$m = \frac{AV}{v}$$

$$m = \frac{A_1 V_1}{v_1} = \frac{A_2 V_2}{v_2} \quad \text{for steady flow}$$

Velocity of fluid leaving the nozzle

- From SFEE

$$h_1 + \frac{V_1^2}{2} + Z_1 g + \frac{\partial q}{\partial m} = h_2 + \frac{V_2^2}{2} + Z_2 g + \frac{\partial w_r}{\partial m}$$

$$h_1 + \frac{V_1^2}{2} = h_2 + \frac{V_2^2}{2}$$

$$V_2 = \sqrt{2(h_1 - h_2) + V_1^2}$$

h in J/kg and **V** in m/s

V1 can be neglected.

$$V_2 = \sqrt{2000(h_1 - h_2) + V_1^2}$$

When **h** is in kJ/kg (from steam Tables)

Critical Pressure ratio

- The value of pressure ratio $\frac{P_2}{P_1}$ at throat

$$\frac{P_2}{P_1} = \left(\frac{2}{n+1} \right)^{\frac{n}{n-1}}$$

$n = 1.3$ for superheated steam

$n = 1.135$ saturated steam

$n = 1.035 + 0.1 x$ for wet steam

Table C.1aSI Saturation Temperature Table for Steam in SI Units

T C	P _{sat} kPa	v _f m ³ /kg	v _g m ³ /kg	v _{fg} m ³ /kg	h _f kJ/kg	h _g kJ/kg	h _{fg} kJ/kg	u _f kJ/kg	u _g kJ/kg	u _{fg} kJ/kg	s _f kJ/kg K	s _g kJ/kg K	s _{fg} kJ/kg K
0	0.6119	0.000995	205.94	205.93	0.9007	2500.02	2499.12	0.9001	2374.02	2373.12	-0.0013	9.1582	9.1595
2	0.7066	0.000995	179.63	179.63	9.2488	2504.40	2495.15	9.2481	2377.48	2368.24	0.0297	9.1052	9.0755
4	0.8140	0.000996	157.04	157.04	17.5909	2508.60	2491.01	17.5901	2380.76	2363.17	0.0604	9.0531	8.9928
6	0.9357	0.000996	137.59	137.59	25.9279	2512.64	2486.72	25.9269	2383.90	2357.97	0.0908	9.0020	8.9113
8	1.0732	0.000997	120.82	120.82	34.2606	2516.58	2482.31	34.2595	2386.91	2352.66	0.1209	8.9519	8.8309
10	1.2282	0.000997	106.31	106.31	42.5897	2520.42	2477.83	42.5885	2389.84	2347.25	0.1508	8.9026	8.7518
12	1.4026	0.000998	93.74	93.74	50.9160	2524.19	2473.27	50.9146	2392.70	2341.79	0.1804	8.8542	8.6738
14	1.5985	0.000999	82.83	82.83	59.2401	2527.90	2468.66	59.2385	2395.51	2336.27	0.2098	8.8066	8.5969
16	1.8180	0.000999	73.33	73.33	67.5625	2531.58	2464.02	67.5607	2398.27	2330.71	0.2389	8.7599	8.5211
18	2.0635	0.001000	65.04	65.04	75.8837	2535.23	2459.34	75.8817	2401.01	2325.13	0.2678	8.7141	8.4463
20	2.3376	0.001000	57.80	57.80	84.2043	2538.85	2454.65	84.2020	2403.73	2319.53	0.2964	8.6690	8.3725
22	2.6431	0.001001	51.47	51.46	92.5247	2542.46	2449.94	92.5220	2406.43	2313.91	0.3249	8.6247	8.2998
24	2.9830	0.001002	45.90	45.90	100.845	2546.06	2445.21	100.842	2409.12	2308.28	0.3531	8.5811	8.2280
26	3.3604	0.001002	41.02	41.02	109.166	2549.65	2440.48	109.163	2411.81	2302.65	0.3811	8.5384	8.1572
28	3.7789	0.001003	36.72	36.71	117.488	2553.23	2435.74	117.484	2414.49	2297.01	0.4090	8.4963	8.0874
30	4.2420	0.001004	32.92	32.92	125.811	2556.81	2431.00	125.807	2417.17	2291.36	0.4366	8.4550	8.0184
32	4.7536	0.001005	29.57	29.56	134.136	2560.39	2426.25	134.131	2419.84	2285.71	0.4640	8.4143	7.9503
34	5.3181	0.001005	26.60	26.60	142.462	2563.96	2421.50	142.456	2422.52	2280.06	0.4913	8.3744	7.8831
36	5.9398	0.001006	23.96	23.96	150.790	2567.53	2416.74	150.784	2425.19	2274.40	0.5183	8.3351	7.8168
38	6.6235	0.001007	21.62	21.62	159.120	2571.09	2411.97	159.113	2427.86	2268.74	0.5452	8.2964	7.7512
40	7.3743	0.001008	19.54	19.54	167.452	2574.65	2407.20	167.444	2430.52	2263.08	0.5719	8.2584	7.6865
42	8.1975	0.001009	17.69	17.69	175.786	2578.20	2402.41	175.778	2433.18	2257.41	0.5985	8.2210	7.6226
44	9.0987	0.001009	16.04	16.03	184.123	2581.75	2397.63	184.114	2435.84	2251.73	0.6248	8.1843	7.5594
46	10.084	0.001010	14.56	14.56	192.463	2585.29	2392.83	192.452	2438.50	2246.04	0.6510	8.1481	7.4970
48	11.160	0.001011	13.23	13.23	200.805	2588.82	2388.02	200.794	2441.14	2240.35	0.6771	8.1125	7.4354
50	12.333	0.001012	12.05	12.04	209.150	2592.34	2383.19	209.137	2443.78	2234.65	0.7030	8.0775	7.3745
52	13.610	0.001013	10.98	10.98	217.498	2595.86	2378.36	217.484	2446.42	2228.93	0.7287	8.0430	7.3143

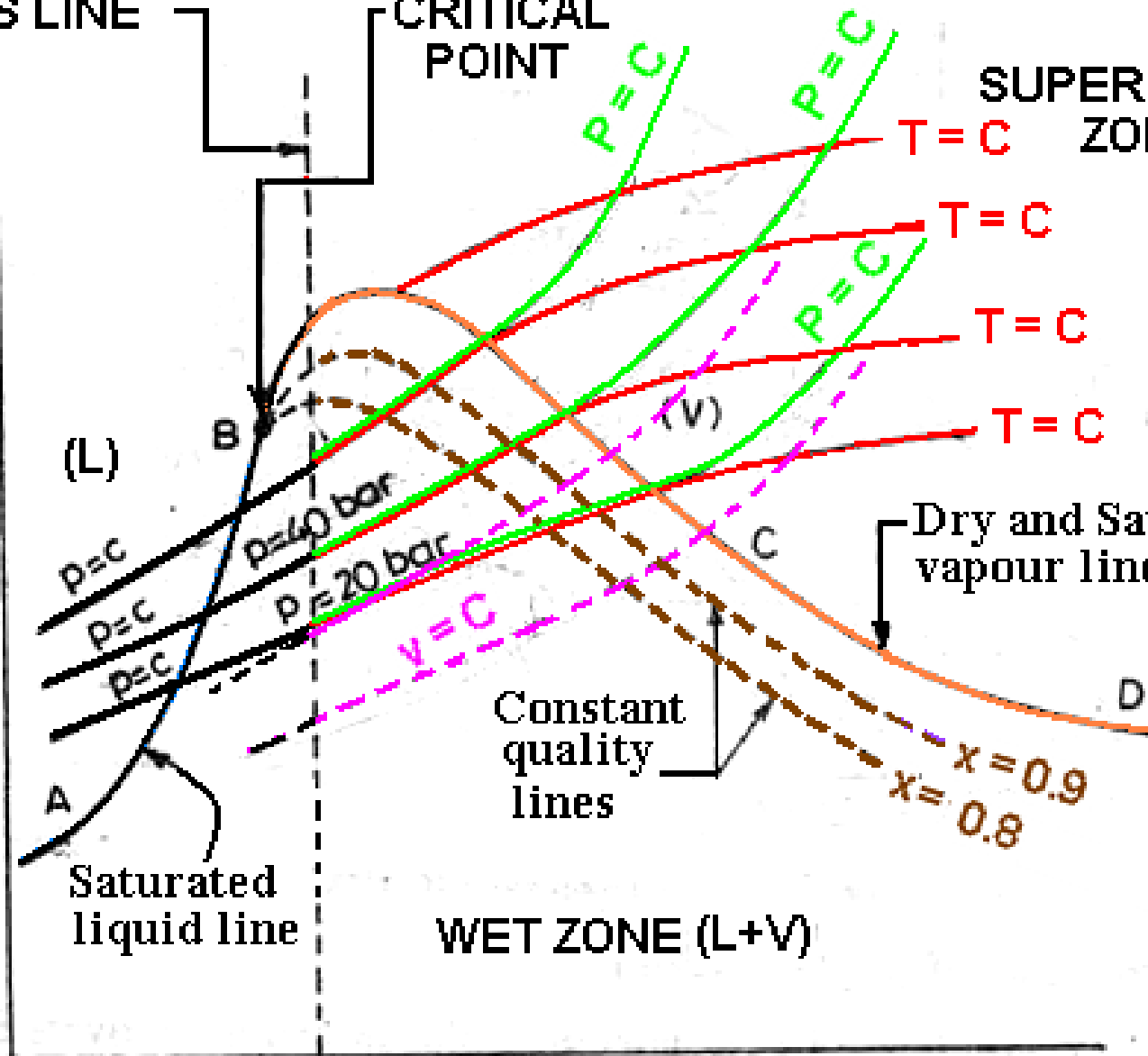
Enthalpy (h), kJ/kg —►

Constant
quality_
lines

-Dry and Saturated vapour line

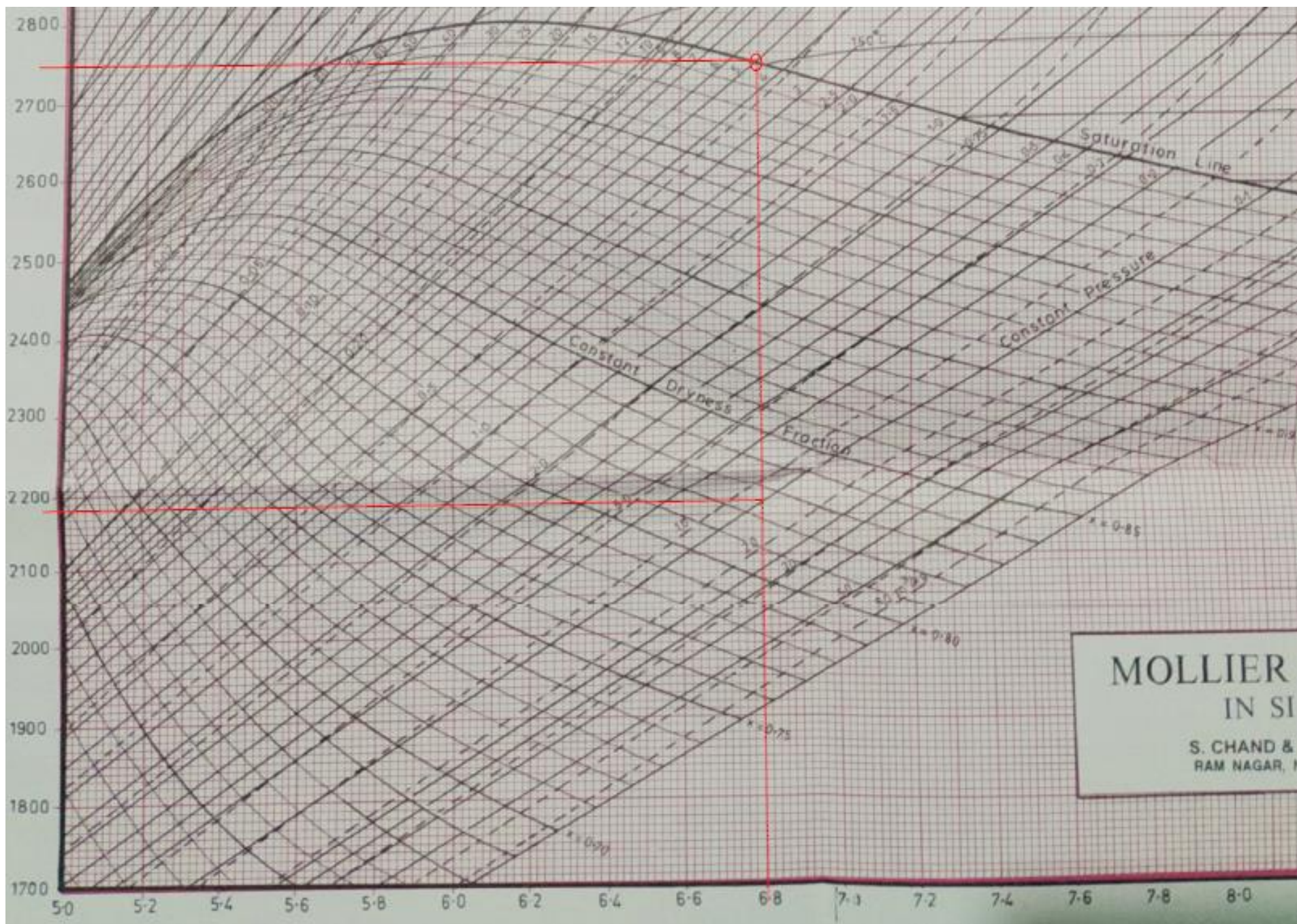
WET ZONE (L+V)

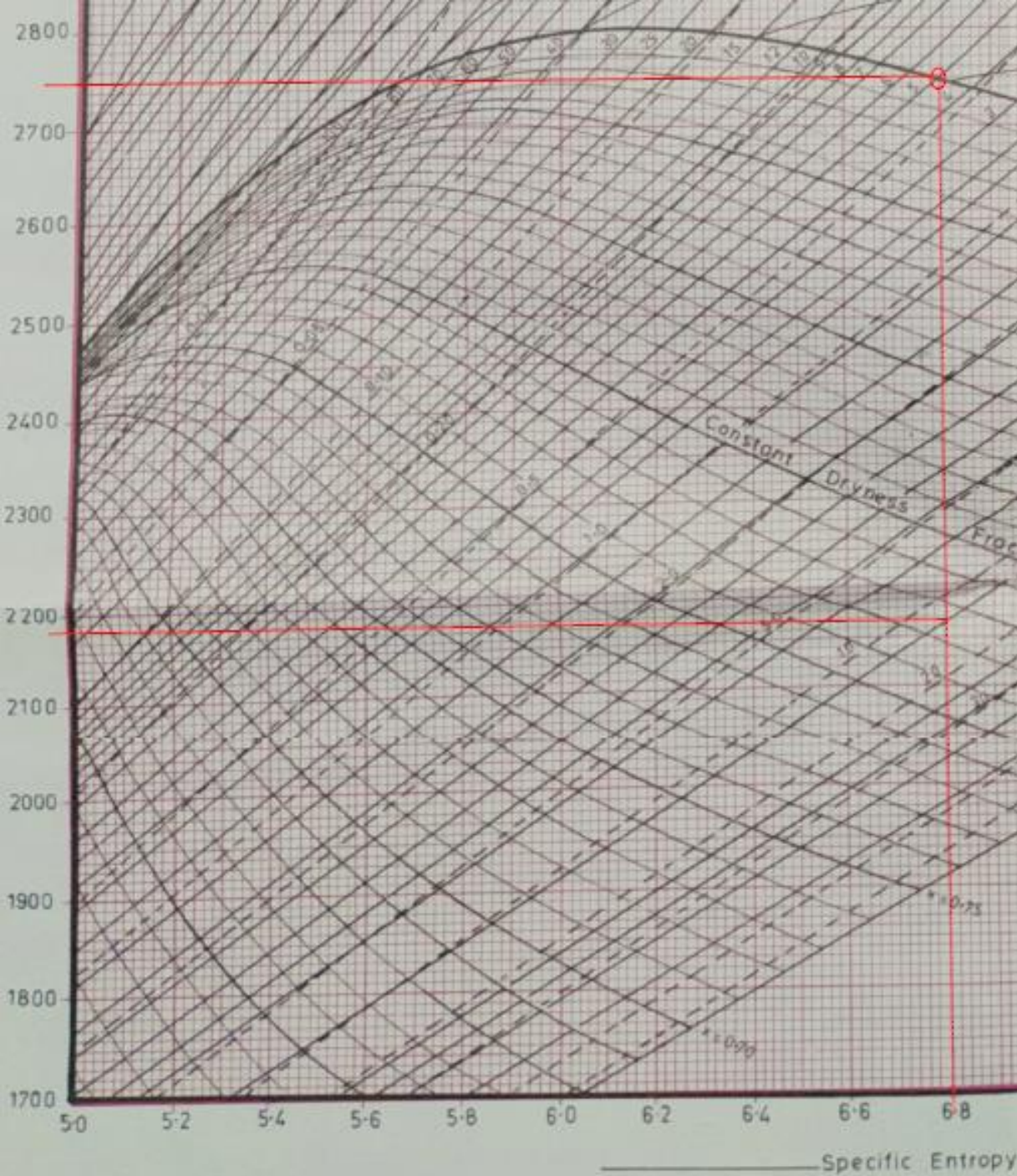
Entropy (s), kJ/kg K $\longrightarrow$



Problem 1

- Steam approaches a nozzle with a velocity of 250 m/s at a pressure of 5 bar dry saturated. If the exit pressure is 0.1 bar, determine the change in enthalpy and exit velocity of the nozzle.





$p_1 = 5 \text{ bar}$

$p_2 = 0.1 \text{ bar}$

$h_1 = 2750 \text{ kJ/kg}$

$h_2 = 2180 \text{ kJ/kg}$

Change in enthalpy = 570 kJ/kg

Exit velocity $V_2 = 1096.58 \text{ m/s}$

V_2

$$= \sqrt{2(h_1 - h_2) \times 1000 + V_1^2}$$

- Steam approaches a nozzle with a velocity of 250 m/s at a pressure of 5 bar dry saturated. If the exit pressure is 0.1 bar, determine the change in enthalpy and exit velocity of the nozzle. Find the exit area of the nozzle if mass flow rate is 0.75 kg/s.

P kPa	T °C	v_f m³/kg	v_g m³/kg	v_{fg} m³/kg	h_f kJ/kg	h_g kJ/kg	h_{fg} kJ/kg	u_f kJ/kg	u_g kJ/kg	u_{fg} kJ/kg	s_f kJ/(kg K)	s_g kJ/(kg K)	s_{fg} kJ/(kg K)
100	99.63	0.001043	1.69	1.69	417.31	2675.27	2257.96	417.20	2505.90	2088.70	1.3020	7.3589	6.0569
125	105.99	0.001048	1.37	1.37	444.14	2685.08	2240.93	444.01	2513.25	2069.24	1.3734	7.2838	5.9104
150	111.37	0.001053	1.16	1.16	466.91	2693.20	2226.29	466.75	2519.34	2052.59	1.4330	7.2225	5.7895
175	116.06	0.001057	1.00	1.00	486.78	2700.13	2213.35	486.59	2524.54	2037.95	1.4843	7.1708	5.6865
200	120.23	0.001061	0.8855	0.8844	504.47	2706.17	2201.70	504.26	2529.07	2024.82	1.5295	7.1260	5.5966
225	123.99	0.001064	0.7930	0.7920	520.46	2711.52	2191.06	520.22	2533.09	2012.86	1.5699	7.0866	5.5167
250	127.43	0.001067	0.7185	0.7175	535.09	2716.32	2181.23	534.83	2536.69	2001.86	1.6065	7.0513	5.4448
275	130.60	0.001070	0.6571	0.6561	548.60	2720.67	2172.07	548.30	2539.95	1991.65	1.6400	7.0194	5.3794
300	133.54	0.001073	0.6057	0.6046	561.16	2724.63	2163.47	560.84	2542.93	1982.09	1.6710	6.9903	5.3193
325	136.29	0.001076	0.5619	0.5608	572.92	2728.28	2155.36	572.57	2545.67	1973.10	1.6997	6.9635	5.2637
350	138.87	0.001079	0.5241	0.5230	583.98	2731.64	2147.66	583.61	2548.20	1964.59	1.7266	6.9386	5.2120
375	141.31	0.001081	0.4913	0.4902	594.44	2734.77	2140.33	594.04	2550.55	1956.51	1.7518	6.9155	5.1637
400	143.62	0.001084	0.4624	0.4613	604.37	2737.68	2133.31	603.93	2552.74	1948.80	1.7756	6.8938	5.1182
425	145.81	0.001086	0.4368	0.4357	613.82	2740.41	2126.59	613.36	2554.79	1941.43	1.7982	6.8735	5.0753
450	147.91	0.001088	0.4139	0.4128	622.85	2742.97	2120.12	622.36	2556.71	1934.36	1.8196	6.8543	5.0347
475	149.91	0.001091	0.3934	0.3923	631.49	2745.37	2113.88	630.97	2558.52	1927.55	1.8400	6.8361	4.9961
500	151.84	0.001093	0.3748	0.3737	639.79	2747.64	2107.86	639.24	2560.23	1920.99	1.8595	6.8188	4.9594
525	153.68	0.001095	0.3580	0.3569	647.77	2749.78	2102.02	647.10	2561.85	1914.75	1.8781	6.8024	4.9243
550	155.46	0.001097	0.3429	0.3418	655.50	2751.80	2096.30	655.10	2563.37	1908.82	1.8958	6.7867	4.8907
575	157.17	0.001099	0.3286	0.3275	662.89	2753.76	2090.87	662.26	2564.83	1902.58	1.9132	6.7717	4.8585
600	158.83	0.001101	0.3156	0.3145	670.07	2755.60	2085.52	669.41	2566.22	1896.80	1.9298	6.7573	4.8275
625	160.43	0.001103	0.3037	0.3026	677.03	2757.35	2080.32	676.34	2567.54	1891.19	1.9458	6.7435	4.7977
650	161.98	0.001105	0.2927	0.2915	683.78	2759.02	2075.24	683.06	2568.79	1885.73	1.9612	6.7302	4.7690
675	163.49	0.001106	0.2824	0.2813	690.33	2760.61	2070.28	689.58	2569.99	1880.41	1.9762	6.7174	4.7412
700	164.95	0.001108	0.2728	0.2717	696.70	2762.13	2065.43	695.92	2571.14	1875.22	1.9906	6.7050	4.7144
725	166.37	0.001110	0.2639	0.2628	702.90	2763.59	2060.69	702.09	2572.24	1870.15	2.0047	6.6931	4.6884
750	167.75	0.001112	0.2556	0.2545	708.93	2764.99	2056.05	708.10	2573.29	1865.19	2.0183	6.6815	4.6632
775	169.09	0.001113	0.2478	0.2467	714.82	2766.33	2051.51	713.96	2574.30	1860.35	2.0315	6.6703	4.6388
800	170.40	0.001115	0.2404	0.2393	720.57	2767.61	2047.05	719.67	2575.27	1855.60	2.0444	6.6595	4.6151

P kPa	T °C	v_f m ³ /kg	v_g m ³ /kg	v_{fg} m ³ /kg	h_f kJ/kg	h_g kJ/kg	h_{fg} kJ/kg	u_f kJ/kg	u_g kJ/kg	u_{fg} kJ/kg	s_f kJ/(kg K)	s_g kJ/(kg K)	s_{fg} kJ/(kg K)
1	6.99	0.000997	128.97	128.97	29.40	2514.61	2485.21	29.40	2385.64	2356.24	0.1058	8.9770	8.8713
2	17.50	0.001000	67.00	67.00	73.42	2534.32	2460.90	73.42	2400.31	2326.89	0.2606	8.7254	8.4648
3	24.09	0.001002	45.66	45.66	100.97	2546.23	2445.26	100.97	2409.25	2308.28	0.3544	8.5791	8.2247
4	28.98	0.001003	34.79	34.79	121.38	2554.99	2433.61	121.38	2415.82	2294.44	0.4225	8.4759	8.0534
5	32.90	0.001005	28.19	28.18	137.74	2561.99	2424.25	137.73	2421.06	2283.33	0.4763	8.3963	7.9200
6	36.19	0.001006	23.73	23.73	151.46	2567.86	2416.40	151.45	2425.46	2274.00	0.5208	8.3314	7.8106
7	39.03	0.001007	20.52	20.52	163.32	2572.92	2409.60	163.32	2429.25	2265.93	0.5590	8.2768	7.7178
8	41.54	0.001008	18.10	18.10	173.80	2577.38	2403.58	173.79	2432.59	2258.79	0.5924	8.2296	7.6372
9	43.79	0.001009	16.20	16.20	183.21	2581.38	2398.17	183.20	2435.58	2252.38	0.6221	8.1881	7.5660
10	45.84	0.001010	14.67	14.67	191.75	2585.00	2393.25	191.74	2438.29	2246.55	0.6489	8.1510	7.5021
15	54.00	0.001014	9.89	9.89	234.15	2591.92	2377.77	234.14	2445.12	2232.20	0.7543	8.0091	7.2547
20	60.09	0.001016	7.75	7.75	264.59	2596.82	2332.23	264.58	2450.68	2205.68	0.8313	7.9090	7.0777
25	64.99	0.001017	6.59	6.59	289.27	2600.61	2311.34	289.26	2454.91	2191.49	0.8924	7.8318	6.9394
30	69.13	0.001018	5.84	5.84	311.98	2603.52	2291.54	311.97	2458.07	2179.48	0.9432	7.7689	6.8257
35	72.71	0.001019	5.29	5.29	332.94	2605.82	2272.87	332.93	2460.11	2169.01	0.9869	7.7159	6.7291
40	75.89	0.001020	4.89	4.89	352.46	2607.65	2255.19	352.45	2461.11	2159.69	1.0252	7.6702	6.6450
45	78.74	0.001021	4.58	4.58	370.45	2609.11	2238.66	370.44	2461.68	2151.27	1.0594	7.6299	6.5705
50	81.35	0.001022	4.34	4.34	387.37	2610.34	2223.97	387.36	2462.30	2143.58	1.0903	7.5939	6.5036
55	83.74	0.001023	4.15	4.15	403.42	2611.32	2210.90	403.41	2462.84	2136.48	1.1185	7.5615	6.4429
60	85.96	0.001024	4.00	4.00	418.73	2612.06	2200.33	418.72	2463.25	2129.89	1.1446	7.5319	6.3873
65	88.02	0.001025	3.89	3.89	433.42	2612.58	2191.16	433.41	2463.55	2123.72	1.1687	7.5047	6.3360
70	89.96	0.001026	3.80	3.80	447.57	2612.91	2183.34	447.56	2463.75	2117.92	1.1912	7.4795	6.2883
75	91.79	0.001027	3.72	3.72	461.25	2613.07	2176.82	461.24	2463.86	2112.43	1.2123	7.4561	6.2438
80	93.51	0.001028	3.65	3.65	474.52	2613.17	2171.65	474.51	2463.89	2107.23	1.2322	7.4343	6.2021
85	95.15	0.001029	3.59	3.59	487.43	2613.22	2167.74	487.42	2463.85	2102.28	1.2510	7.4138	6.1628
90	96.72	0.001030	3.54	3.54	499.99	2613.23	2165.09	499.98	2463.75	2097.56	1.2688	7.3945	6.1256
95	98.21	0.001031	3.50	3.50	512.29	2613.20	2163.91	512.28	2463.60	2093.04	1.2858	7.3762	6.0904

$h_f = 191.75$ kJ/kg,

$s_f = 0.6489$ kJ/kgK

$v_f = 0.001010$ m³/kg

$h_{fg} = 2393.25$ kJ/kg

$s_{fg} = 7.5021$ kJ/kgK

$v_{fg} = 14.67$ m³/kg

- Steam approaches a nozzle with a velocity of 250 m/s at a pressure of 5 bar dry saturated. If the exit pressure is 0.1 bar, determine the change in enthalpy and exit velocity of the nozzle. Find the exit area of the nozzle if mass flow rate is 0.75 kg/s.

At 5 bar (500 kPa)

$$h_1 = h_g = 2747.64 \text{ kJ/kg} \quad s_1 = s_g = 6.8188 \text{ kJ/kgK}$$

At 0.1 bar (10 kPa)

$$\begin{array}{ll} h_f = 191.75 \text{ kJ/kg}, & h_{fg} = 2393.25 \text{ kJ/kg}, \\ s_f = 0.6489 \text{ kJ/kgK} & s_{fg} = 7.5021 \text{ kJ/kgK} \\ v_f = 0.001010 \text{ m}^3/\text{kg} & v_{fg} = 14.67 \text{ m}^3/\text{kg} \end{array}$$

$$s_1 = s_2 \quad s_2 = s_f + x \cdot s_{fg}$$

$$6.8188 = 0.6489 + x \cdot 7.5021$$

$$x = 0.822$$

$$h_2 = h_f + x \cdot h_{fg} = 191.75 + 0.822 \times 2393.25 = 2159 \text{ kJ/kg}$$

$$v_2 = v_f + x \cdot v_{fg} = 0.001010 + 0.822 \times 14.67 = 12.06 \text{ m}^3/\text{kg}$$

$$V_2 = \sqrt{2(h_1 - h_2) \times 1000 + V_1^2}$$

$$V_2 = \sqrt{2(2747.64 - 2159) \times 1000 + 250^2}$$

$$V_2 = 1113.45 \text{ m/s}$$

- Steam approaches a nozzle with a velocity of 250 m/s at a pressure of 5 bar dry saturated. If the exit pressure is 0.1 bar, determine the change in enthalpy and exit velocity of the nozzle. Find the exit area of the nozzle if mass flow rate is 0.75 kg/s.

$$s_1 = s_2 \qquad s_2 = s_f + x \cdot s_{fg}$$

$$6.8188 = 0.6489 + x \cdot 7.5021$$

$$x = 0.822$$

$$h_2 = h_f + x \cdot h_{fg} = 191.75 + 0.822 \times 2393.25 = 2159 \text{ kJ/kg}$$

$$v_2 = v_f + x \cdot v_{fg} = 0.001010 + 0.822 \times 14.67 = 12.06 \text{ m}^3/\text{kg}$$

$$V_2 = \sqrt{2(h_1 - h_2) \times 1000 + V_1^2}$$

$$V_2 = \sqrt{2(2747.64 - 2159) \times 1000 + 250^2}$$

$$V_2 = 1113.45 \text{ m/s}$$

$$m = \frac{A_1 V_1}{v_1} = \frac{A_2 V_2}{v_2}$$

$$0.75 = \frac{A_2 \times 1113.45}{12.06}$$

$$A_2 = 8.1234 \times 10^{-3} \text{ m}^2 = 81.234 \text{ cm}^2$$

Determine the mass flow rate of steam through a nozzle having isentropic flow through it. Steam enters nozzle at 10 bar, 500 °C and leaves at 6 bar. Cross-section area at exit of nozzle is 20 cm². Velocity of steam entering nozzle may be considered negligible. Show the process on h-s diagram also.

$$h_1 = 3498 \text{ kJ/kg,}$$

$$h_2 = 3308 \text{ kJ/kg K}$$

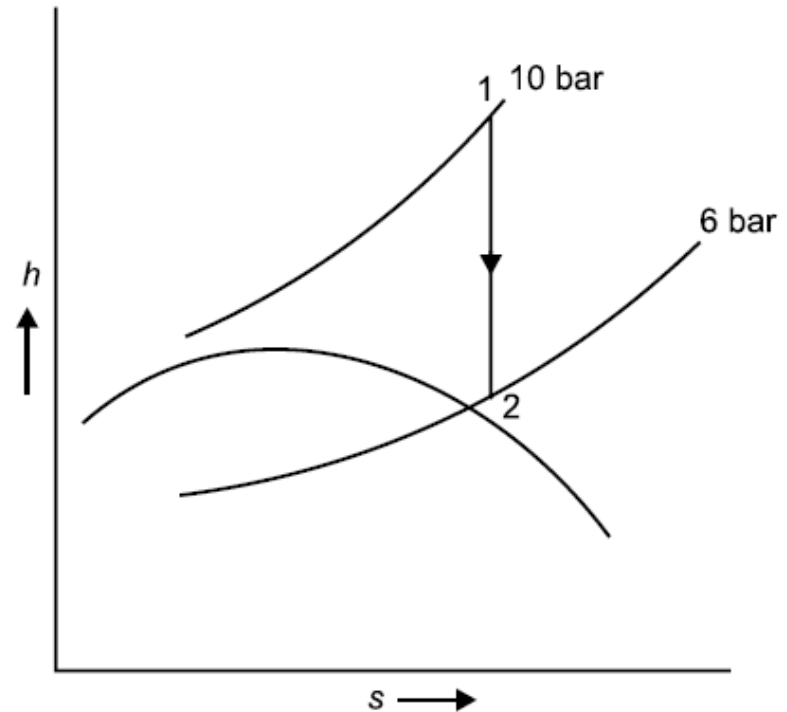
$$V_2 = \sqrt{2 \times (h_1 - h_2)}$$

$$= 616.44 \text{ m/s}$$

$$m = \frac{A_2 \times V_2}{v_2}$$

$$v_2 = 0.55 \text{ m}^3/\text{kg}$$

$$\text{Mass flow rate} = 2.24 \text{ kg/s}$$



$$h_1 = 3498 \text{ kJ/kg,}$$

$$h_2 = 3308 \text{ kJ/kg K}$$

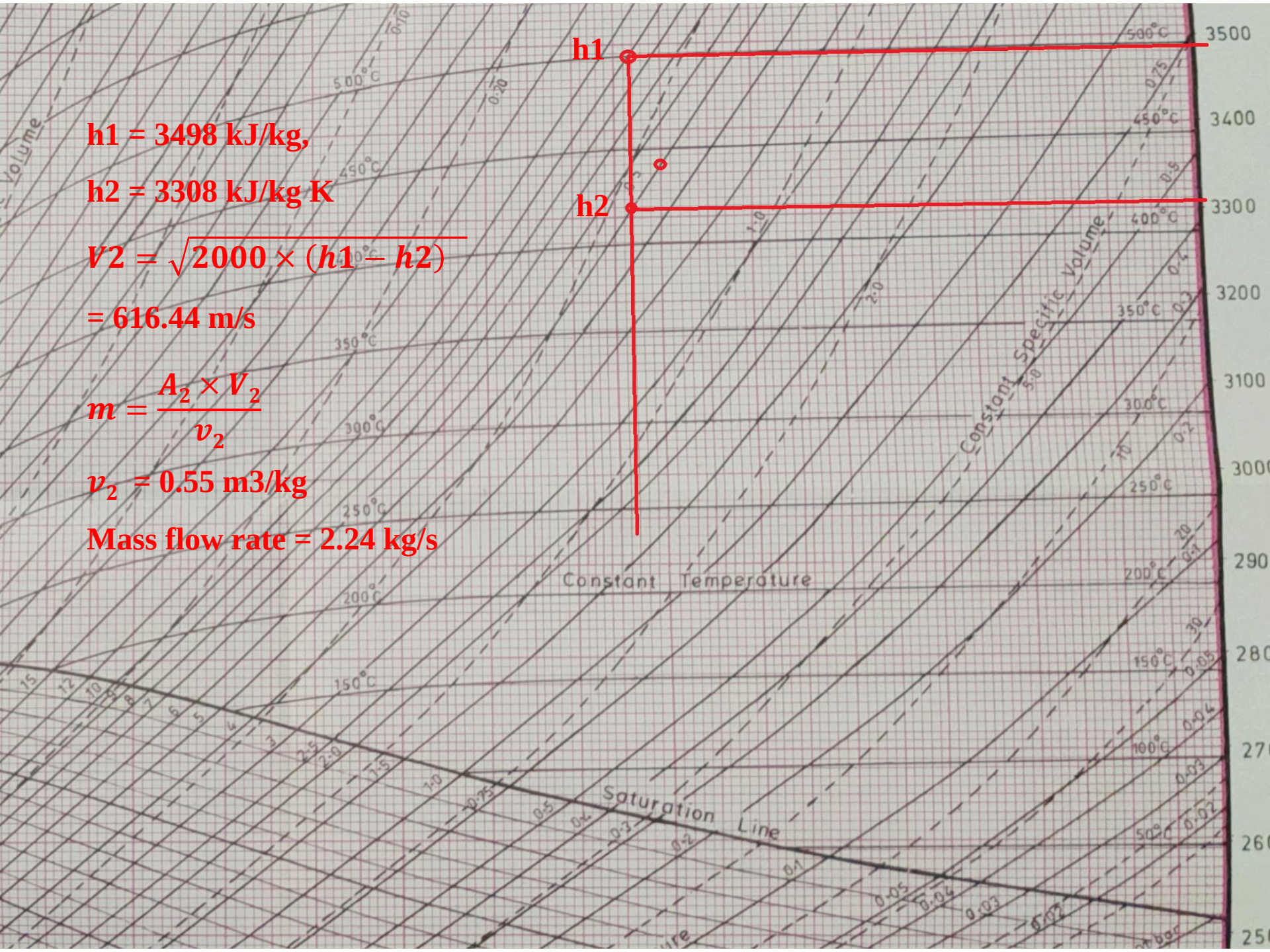
$$V_2 = \sqrt{2000 \times (h_1 - h_2)}$$

$$= 616.44 \text{ m/s}$$

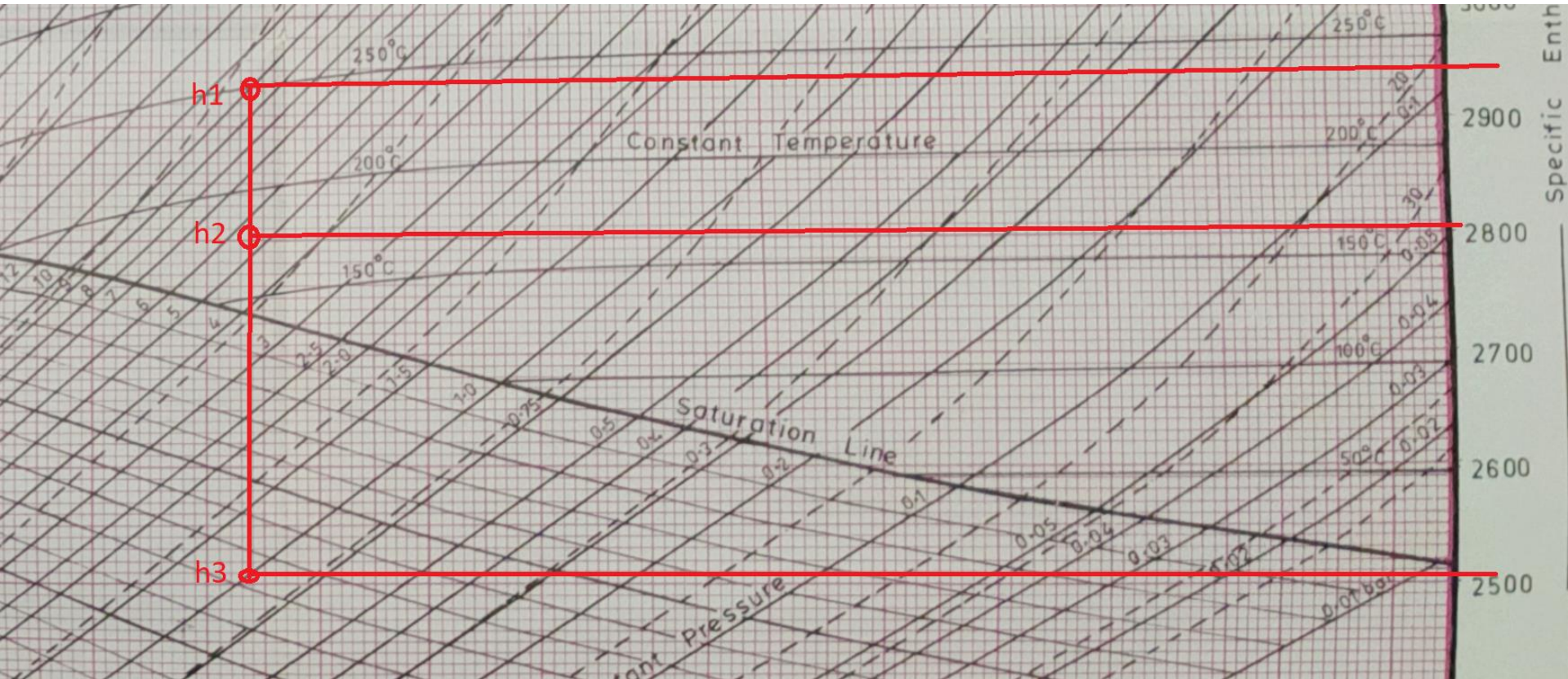
$$m = \frac{A_2 \times V_2}{v_2}$$

$$v_2 = 0.55 \text{ m}^3/\text{kg}$$

$$\text{Mass flow rate} = 2.24 \text{ kg/s}$$



P3: Steam expands from 10 bar 250 °C to 1 bar.
Determine throat velocity and exit velocity.



$h_1 = 2940 \text{ kJ/kg}$, $h_3 = 2508 \text{ kJ/kg}$

$V_1 = 0 \text{ m/s}$

- Steam expands from 10 bar 250 °C to 1 bar. Determine throat velocity and exit velocity.

The value of pressure ratio $\frac{P_2}{P_1}$ at throat

$$\frac{P_2}{P_1} = \left(\frac{2}{n+1} \right)^{\frac{n}{n-1}}$$

$n = 1.3$ for superheated steam

$n = 1.135$ saturated steam

$n = 1.035 + 0.1 \times$ for wet steam

$$\frac{P_2}{10} = \left(\frac{2}{1.3+1} \right)^{\frac{1.3}{1.3-1}} = 0.546$$

$$P_2 = 5.46 \text{ bar} \cong 5.5 \text{ bar}$$

$$h_2 = 2805 \text{ kJ/kg}$$

$$h_1 = 2940 \text{ kJ/kg}, \quad h_3 = 2508 \text{ kJ/kg}, \quad V_1 = 0$$

$$\text{Throat Velocity } V_2 = \sqrt{2(h_1 - h_2) \times 1000 + V_1^2}$$

$$V_2 = \sqrt{2(2940 - 2805) \times 1000} = 519.61 \text{ m/s}$$

$$\text{Exit Velocity } V_3 = \sqrt{2(h_1 - h_3) \times 1000 + V_1^2}$$

$$V_3 = \sqrt{2(2940 - 2508) \times 1000}$$

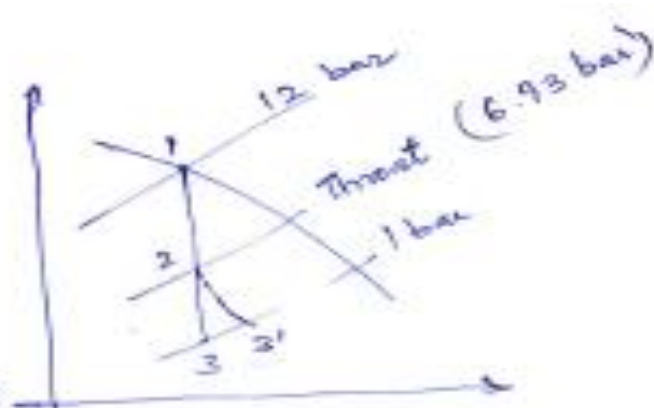
$$V_3 = 929.5 \text{ m/s}$$

Dry saturated steam at a pressure of 12 bar flows through the nozzles at the rate of 4.5 kg/s and discharge at a pressure of 1 bar. The loss due to friction occurs only in the divergent portion of the nozzle and the magnitude is 10% of the total isentropic enthalpy drop. Determine the cross sectional area at the throat and exit of the nozzle.

Critical pressure ratio

$$\frac{P_2}{P_1} = \left(\frac{2}{n+1} \right)^{\frac{n}{n-1}}$$

where n isentropic expansion index



- $n = 1.3$ for super heated steam
- $n = 1.135$ for Dry saturated steam
- $n = 1.035 + 0.1x$ for wet steam (Zeuner's eqn)

Remember this

$$\frac{P_2}{P_1} = \left(\frac{2}{1.135+1} \right)^{\frac{1.135}{0.135}} = 0.5774$$

$$\therefore P_2 = 0.5774 \times 12 = 6.93 \text{ bar}$$

$$h_1 = 2782.7 \text{ kJ/kg} \quad h_2 = ?$$

$$s_1 = 6.5194 = s_3$$

$$s_3 = 1.3027 + x(7.3598 - 1.3027)$$

$$x_3 = 0.86$$

$$h_2 = h_f + x h_{fg} = 417.51 + 0.86(2675.4 - 417.51) = 2359.29 \text{ kJ/kg}$$

$$h_1 - h_3 = 423.41 \text{ kJ/kg}$$

$$h_1 - h_{3'} = 0.9 \times 423.41 = 381.07 \text{ kJ/kg}$$

$$h_{3'} = 2401.62 \text{ kJ/kg}$$

$$h_2 =$$

$$\alpha_2$$

$$S_2 = S_f + \alpha \cdot S_{fg} @ 6.93 \text{ bar}$$

$$\alpha_p = 0.9606$$

$$h_2 = 697.06 + 0.9606 \times 2064.9, \quad v_2 = 0.0011022 + 0.9606 \times (0.27218 - 0.0011022)$$

$$= 2680.60 \text{ kJ/kg} \quad \quad \quad = 0.26198 \text{ m}^3/\text{kg}$$

$$V_2 = \sqrt{2(h_1 - h_2)} = \sqrt{2(2782 - 2680.6) \times 1000} = 450.33 \text{ m/s}$$

$$V_3' = \sqrt{2(h_1 - h_3')} = \sqrt{2 \times 381.07 \times 1000} = 873.01 \text{ m/s}$$

$$m = \frac{A_2 V_2}{v_2} \quad A_2 = \frac{m v_2}{V_2} = \frac{4.5 \times 0.26198}{450.33} = 0.002618 \text{ m}^2$$

$$= 26.18 \text{ cm}^2$$

$$A_3' = \frac{m v_3'}{V_3'}$$

$$\alpha_3 = 0.8787$$

$$h_3' = 0.0010434 + 0.8787 \times 1.6937$$

$$= 1.489 \text{ m}^3/\text{kg}$$

$$A_3 = \frac{4.5 \times 1.489}{873.01} = 0.007675 \text{ m}^2 = 76.75 \text{ cm}^2$$

Steam at a pressure of 10 bar and 95% dry is supplied through a convergent-divergent nozzle to the wheel chamber, where the pressure is maintained at 0.12 bar. The mass flow rate through the nozzle is 8.16 kg/kW-hr and the work developed by the wheel is 110 kW. Determine:

- pressure at the throat using Zeuner's equation
- no. of nozzles needed if each nozzle has a throat diameter of 5 mm.
- the exit diameter of the nozzle if 10% of the overall enthalpy drop overcomes friction in the divergent portion resulting in the divergent portion.

By Zeuner's eqn., $n = 1.035 + 0.1 x$

$$n = 1.035 + 0.1 \times 0.95 = 1.13$$

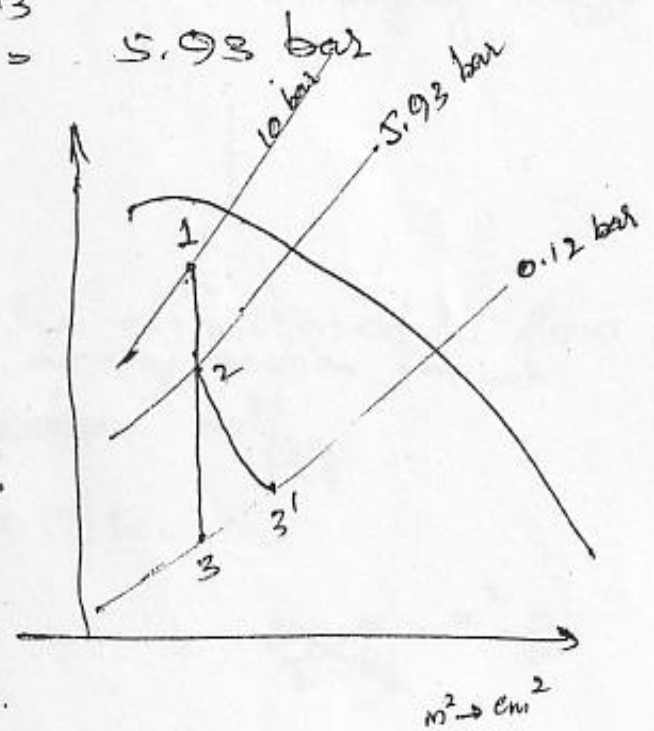
$$P_2 = P_1 \cdot \left(\frac{2}{n+1} \right)^{\frac{n}{n-1}} = 10 \times \left(\frac{2}{1.13+1} \right)^{\frac{1.13}{0.13}} = 5.93 \text{ bar}$$

$$\eta_{\text{nozzle}} = 90\% \quad \text{or} \quad 0.9 = \frac{h_1 - h_3'}{h_1 - h_3}$$

$$m = \frac{8.16 \times 110}{3600} = 0.25 \text{ kg/sec}$$

$$V_2 = \sqrt{2 \times 1000 (h_1 - h_2)}$$

$$= \sqrt{2 \times 1000 \times (2675 - 2580)} = 436 \text{ m/sec.}$$



Combined area of throat $A_2 = \frac{m V_2}{V_2} = \frac{0.25 \times 0.305 \times 10^4}{436} = 1.74 \text{ cm}^2$

Throat area per nozzle $= \frac{\pi}{4} (0.5)^2 \text{ cm}^2 = 0.1965 \text{ cm}^2$

No. of nozzles $= \frac{1.74}{0.1965} = 8.85 = \underline{\underline{9}}$

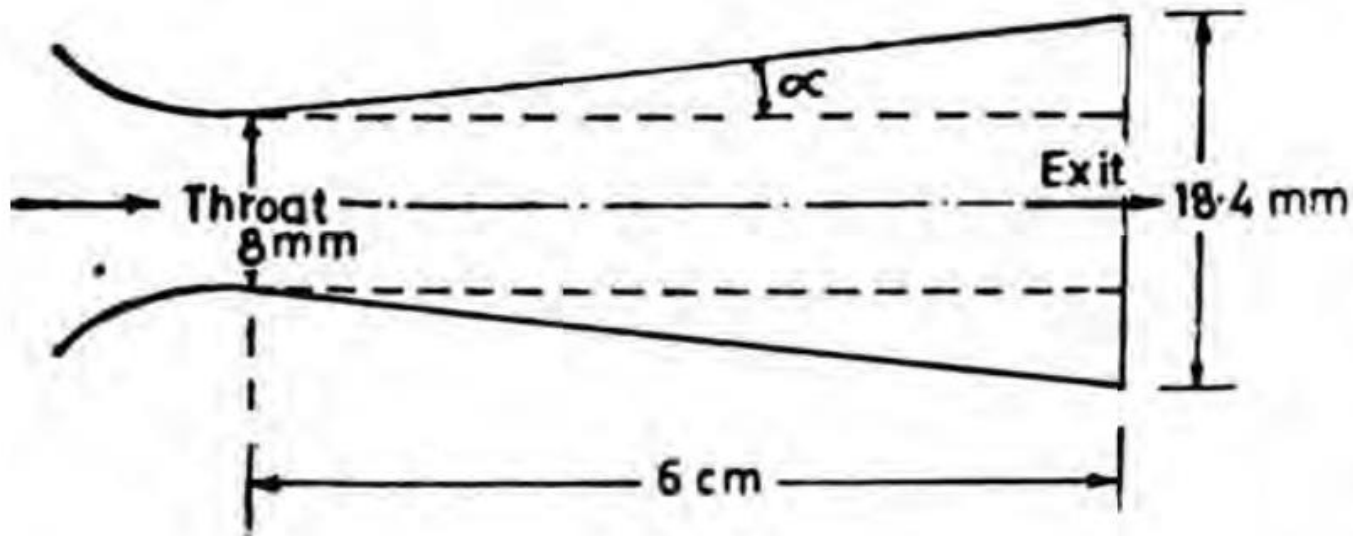
$$\text{Nozzle exit velocity } V_3' = \sqrt{2 \times 1000 (h_1 - h_3')} = \sqrt{2 \times 1000 \times 575} = 1072 \text{ m/s.}$$

$$\text{Combined area of exit} = \frac{m V_3'}{V_3'} = \frac{0.25 \times 10 \times 10^4}{1072} = 23.32 \text{ cm}^2$$

$$\text{Area of 1 nozzle} = \frac{23.32}{9} = 2.59 \text{ cm}^2$$

$$\therefore \text{Nozzle dia at exit} = \sqrt{\frac{2.59 \times 4}{\pi}} = \underline{\underline{1.81 \text{ cm}}}$$

- A CD nozzle is supplied with dry saturated steam at 11 bar. The divergent portion of the nozzle is 6 cm long and the throat diameter is 8 mm, determine the semi-cone angle of the divergent part of the nozzle so that the steam may leave the nozzle at 0.4 bar. Neglect friction in the nozzle.



- A CD nozzle is supplied with dry saturated steam at 11 bar. The divergent portion of the nozzle is 6 cm long and the throat diameter is 8 mm, determine the semi-cone angle of the divergent part of the nozzle so that the steam may leave the nozzle at 0.4 bar. Neglect friction in the nozzle.

$$\tan \alpha = \frac{D_3 - D_2}{2l}$$

all dimensions in mm

$$m = \frac{A_2 V_2}{v_2}$$

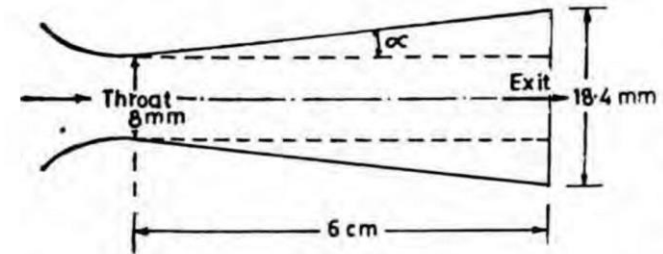
Find m

Find h_1 , s_1 , x_2 , h_2 , x_3 , h_3

Find V_2

$$\frac{A_3 V_3}{v_3} = \frac{A_2 V_2}{v_2}$$

Find V_3 , A_3 and D_3



- In a steam nozzle steam expands from 16 bar to 5 bar with initial temperature of 300 °C and mass flow of 1 kg/s. Determine the throat and exit areas considering (i) expansion to be frictionless and, (ii) friction loss of 10% throughout the nozzle.

$$h_1 = 3034.8 \text{ kJ/kg at 16 bar, 300C}$$

$$s_1 = 6.8844 \text{ kJ/kgK}$$

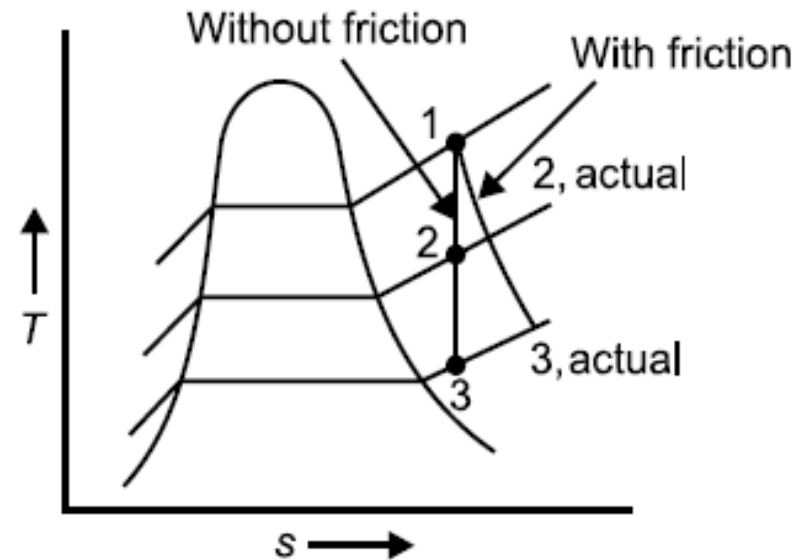
$$v_1 = 0.15862 \text{ m}^3/\text{kg}$$

$$h_2 = 2891.39 \text{ kJ/kg}$$

$$h_3 = 2777.0 \text{ kJ/kg}$$

$$v_2 = 0.2559 \text{ m}^3/\text{kg}$$

$$v_3 = 0.3882 \text{ m}^3/\text{kg}$$



- $V2 = \sqrt{2 \times (h1 - h2)}$

- $V2 = 539.9 \text{ m/s}$

- $A2 = 3.209 \times 10^{-3} \text{ m}^2$

- $A3 = 3.647 \times 10^{-3} \text{ m}^2$

An impulse turbine of 1 MW has steam entering at 20 bar and 300 °C and steam consumption of 8 kg per kW hour. Steam leaves at 0.2 bar and 10% of total heat drop is lost in overcoming friction in diverging portion of nozzle. If throat diameter of each nozzle is 1 cm then determine (i) the number of nozzles required (ii) exit diameter of each nozzle.

